

Screen Sense Kids' Screentime Visualization

Milestone 04 Report (Week 7 & Week 8)

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OBJECTIVE

The objective of Milestone-4 is to transform analytical insights obtained from Milestones 1, 2, and 3 into a professional, interactive dashboard and final presentation. This milestone focuses on presenting screen-time patterns in a clear and understandable manner for parents, teachers, and non-technical stakeholders, ensuring that insights are actionable and visually intuitive.

Week 7: Visual Report / Dashboard Preparation

Dashboard Design Approach:

An interactive Power BI dashboard was developed by integrating insights from:

- Exploratory analysis (Milestone-2)
- Cohort and habit analysis (Milestone-3)

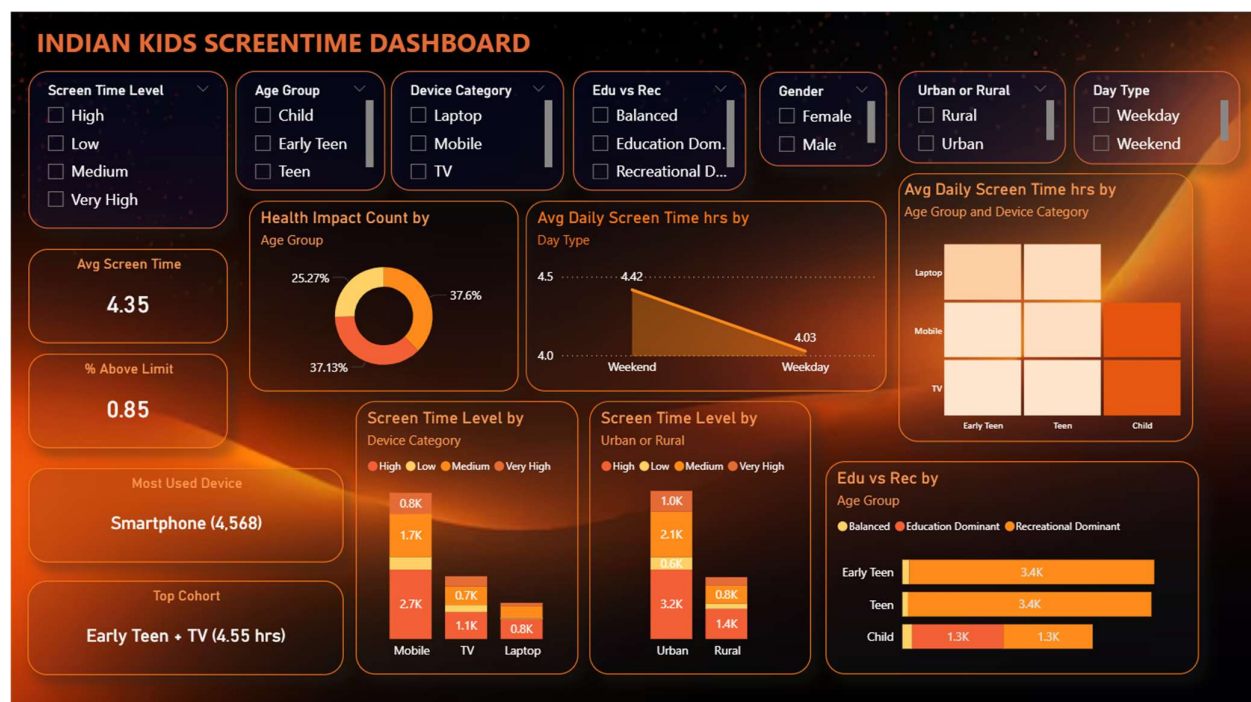
Special focus was given to:

- Visual clarity
- Minimal clutter
- Logical storytelling
- Controlled slicer interactions

The dashboard follows a top-down narrative structure:

1. Overall screen-time risk (KPIs)
2. Cohort and segment breakdowns
3. Habit-based behavior insights

Dashboard:



Dashboard Components

KPI Cards

The following KPI indicators summarize key behavioural risks:

- Average Screen Time (hrs): Shows overall usage severity.
 - % Above Safe Limit: Indicates proportion exceeding recommended screen time.
 - Most Used Device (with count): Identifies dominant device dependency.
 - Top Risk Cohort (Age + Device): Highlights the most vulnerable group.
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Cohort & Segment Analysis Visuals (Week-5 Insights)

- Average Daily Screen Time Hrs by Age Group & Device Category (Heatmap):
Identifies which age-device combinations contribute most to high screen-time usage.
 - Screen Time Level × Urban or Rural (Stacked Column):
Shows how screen-time intensity differs by location.
 - Edu vs Rec × Age Group (Stacked Bar):
Explains whether screen usage is education-driven or entertainment-driven across age groups.
 - Health Impact Count × Age Group (Donut Chart):
Analyses how reported health impacts vary across different age groups.
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Habit Pattern Visuals (Week-6 Insights)

- Weekday vs Weekend Average Screen Time (Line Chart):
Reveals time-based screen-usage habits.
 - Screen Time Level × Device Category (Stacked Column):
Shows which devices dominate low, Medium, High, Very High screen-time categories.
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Interactive Slicers

The dashboard includes the following slicers:

- Age Group
 - Gender
 - Device Category
 - Screen Time Level
 - Day Type
 - Urban or Rural
 - Educational vs Recreational
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Week 8: Documentation and Final Presentation

Description

Week 8 focused on documenting the project and preparing a structured presentation of the Dashboard. A final report and slide deck were created to explain:

- Dataset understanding
- Analytical workflow
- Dashboard design
- Key insights and findings

The presentation includes a walkthrough of dashboard interactions and explains how slicers and KPIs work together to reveal meaningful insights.

Key Insights from the Dashboard

- Early Teens and Teens form the highest screen-time cohort.
- Mobile devices dominate high and very-high screen-time levels.
- Weekend screen time is significantly higher than weekdays.
- Urban children exhibit higher screen-time risk than rural children.
- Recreational usage drives excessive screen-time behavior.
- A large percentage of users exceed recommended daily screen-time limits, highlighting digital well-being concerns.

Conclusion

Milestone 4 successfully brings together all analytical work into a unified dashboard and presentation. The final dashboard provides actionable insights into children's screen-time behaviours by identifying high-risk groups, device dependency, and usage habits. This project demonstrates how data visualization can support parents, educators, and policymakers in understanding and addressing excessive screen-time usage among children.